viii) Environment (Protection) Act, 1986 (B) [Ans: 4:04]

10. Write a note on the following :
    (a) Central Pollution Control Board (C) [Ans : 4:07 + 4:07:1]
    (b) State Pollution Control Board (C) [Ans: 4:08 + 4:08:1]

11. What are the powers of the State Pollution Control Board? (B) [Ans: Points 1 to 6 under 4:08:2]

12. Discuss briefly about the organizations established to ensure environmental protection in India. (A) [Ans: 4:05 + 4:06 + 4:06:1 + 4:06:02 + 4:07 + 4:07:1 + 4:08 + 4:08:1+ 4:08:2]

13. Write a short note on the following :
    i) Environmental Foundation of Africa (B) [Ans. 4:09:1 + 4:09:1:01]
    ii) Worldwide Fund for Nature (B) [Ans. In 4:09:2 first 5 lines (ending with environment) + 4:09:2:02]
    iii) Conservation International (B) [Ans. 4:09:3 + 4:09:3:01]
    iv) International Union for Conservation of Nature (B) [Ans. 4:09:5:01 + 4:09:5:02]
    v) Aims of Greenpeace (B) [Ans. 4:09:4:01]
    vi) Major achievements of Greenpeace (B) [Ans. 4:09:4:02]
    vii) Efforts taken in India to ensure environmental protection (C) [Ans. 4:05]

# Unit - V

# ENVIRONMENTAL EDUCATION IN THE SCHOOL CURRICULUM

## 5:0 Introduction

Environmental Education starts in a child's home and immediate neighbourhood. A child's perception of the environment develops both at home and through formal schooling in Nursery schools and Pre-primary Institutions. According to **Hague**, 'environmental study is an approach through activities based on child's physical and social environment which leads to the progressive development of attitude and skills required for the observation, recording, interpretation and communication of scientific, historical and geographical data'. The NCERT developed a curriculum for environmental education. The chief objective of environmental education is to give the students, experience of an approach to learning by inquiry.

The International Environmental Education Programme (IEEP) which began in 1975 has urged the inclusion of environmental issues in national curricula. About 60 countries have introduced environmental education and taken into their school plans. The primary aim of environmental education is to enable human beings to understand the complex nature of the environment as the results from the interaction of its biological, physical, social and cultural aspects. Children should be given direct experiences in the real environment, so that they may develop an awareness and appreciation of the factors and interrelationships operating in the environment, leading to an ecological understanding of man's oneness with his environment'.

In this unit, status of environmental education in school curriculum, incorporating environmental education in the school curriculum at different levels of formal education, innovative methods of teaching environmental education, problems faced in teaching environmental education, role of UNEP, CEE and NCERT in promoting environmental education are taken for discussion.

## 5:01 Environmental Education in National Policy on Education (1986)

National policy on Education (NPE), 1986 has recommended core areas that are common for all subjects of study. One of the common cores listed in



the policy document is the protection of environment. The NCERT developed a National Curriculum for Elementary and Secondary education: 'A Framework (NCF)' in 1988 to incorporate the major thrusts and recommendations of NPE. Among other aspects the NCF also elaborates the need for developing environmental consciousness and preparing children for environmental protection through education.

NCF recommended that at the lower primary level an integrated approach be followed. In grades I and II, the child is to be introduced to the environment as a whole without making any distinction between natural and social elements; from grade III onward the environmental focus continues. The physical and social aspects of the environment are to be introduced respectively as environmental studies (science) and social studies as a broad and composite area of study. In view of the policy recommendation for providing science education for all, a composite course of 'science' has been extended from upper primary to secondary stage. The curriculum of general education is quite rich with EE dimensions.

## 5:02 Need for Environmental Education in the School Curriculum

The general needs for inclusion of environmental education in school curriculum are as follows :

i) To create awareness about various environmental issues among the school students.

ii) To make them environmentally literate. By acquiring environmental awareness, they can make proper decisions to protect the environment.

iii) To provide necessary attitude for the conservation of environment among the school students.

iv) To provide the knowledge about ecological systems and 'cause and effect' relationship.

v) To understand the ecological energy flow system and its influence on man.

vi) To develop skill to solve the environmental problems among the school learners.

vii) To provide necessary information about bio-diversity richness and the potential dangers to the species of plants, animals and microorganisms.



viii) To understand the causes and consequences due to natural and man induced disasters.

ix) To study the problems of over population, health, hygiene etc., and to find out the way of eliminating these problems.

x) To realize the need for sustainable utilization of natural resources.

xi) To develop an appreciation of species and habitat management techniques.

xii) To understand the influences of humans on the environment.

## 5:03 Providing Environmental Education at Different Levels of Education

Materials for environmental education are still in an involving stage. The six key characteristics of good environmental education material are: fairness and accuracy, depth, emphasis on skill building, action orientation, instructional soundness and usability. In India, the course content of environmental education can be classified stage wise. These stages are :

- **Primary Education Stage,** where the emphasis should be largely on building environmental awareness followed by real life situations and conservation. As the attempt at this stage is to sensitize the child to the environment, the content to be used includes the child's surroundings.

- **Lower Secondary Stage,** where the quantum of awareness decreases with increased knowledge of real life situations, conservation and sustainable development. As the objectives at this stage are real life experiences, awareness and problem identification, the contents to be used are the subjects taught at the primary school level supplemented with general science and field visits.
- **Higher Secondary Stage,** where the emphasis must be on conservation, assimilation of knowledge, problem identification and action skills. The content used in this phase may be based on science with practical work and field work.
- **College Stage,** where the emphasis must be on sustainable development, followed by conservation, real life situations and problem-solving. The content must be based on science and technology.



## 5:04 Approaches in Providing Environmental Education

The approaches to Environmental Education can be categorized into two classes viz. ***Interdisciplinary Approach and Multidisciplinary Approach.***

### 5:04:1 Interdisciplinary Approach

In this approach Environmental Education is given a separate identity. Topics are culled from various subjects like social studies, science, mathematics etc., and are made as an independent subject of study.

### 5:04:2 Multidisciplinary Approach

In this approach Environmental Education is not given a separate identity. Issues relating to environment are included in the other conventional subjects.

Among the two approaches viz (i) Inter-Disciplinary Approach and (ii) Multi-Disciplinary Approach, the latter is being preferred in schools as the teachers handling other subjects can teach components of Environmental Education too during their classes, avoiding the need for a separate teacher of Environmental Education.

A few suggestions for integrating EE with the regular school subjects under Multi-Disciplinary approach are given below :

**Chemistry :**

**Water** : Pollution - sources of pollution - reasons - population explosion - deforestation - over-exploitation of groundwater - water scarcity.

**Air** : Air pollution - sources - effects - importance of trees in the production of oxygen - ozone layer depletion - acid rain-greenhouse effect - relation between population and pollution - importance of nitrogen cycle - air pollutants from industries.

**Botany** : Importance of trees in preventing pollution - ozone depletion - noise control - trees in preventing soil erosion - use of medicinal plants - role of trees in building economy - use of pesticides - natural way of controlling pests.

**Zoology** : Protection and preservation of wildlife - evolution - extinction of species - need for conservation - food chain and food - web - need for bio-diversity - balance of nature.



**Physics** : Conservation of energy - alternative sources of energy - solar cookers - biogas plants - windmills - noise pollution - trees reducing noise pollution -nuclear energy - harmful effects of radiation.

**Geography** : Importance of forest areas - mineral survey - CFC - river systems - desertification.

**History** : The rise and fall of civilisations - wars and hazards of war - nuclear warfare.

## 5:05 Approaches in Incorporating Environmental Education in the Curriculum

The Approaches of Environmental Education can also be categorized in the following ways on the basis of incorporating Environmental Education in the curriculum:

- Environmental Education as a separate subject
- Infusing Environmental Education in the Existing Subjects
- Occasional Programmes

- Environmental Education as Core Curriculum.

### 5:05:1 Environmental Education as a Separate Subject

In this approach, Environmental Education is treated as a separate subject drawing the basic principles and skills from other conventional disciplines. In other words, Environmental Education becomes another subject on par with other established subjects like mathematics, science etc. In this approach, Environmental Education is accorded the same priority as other subjects. Another advantage of this approach is that it is possible to develop a holistic picture of a body of knowledge, which may help its transfer to everyday behaviour. The disadvantage of this approach is that it adds to the existing heavy curriculum. While developing the curriculum in this approach, some of these assumptions are made :

1) All children in all schools need Environmental Education
2) Environmental Education curriculum in each grade
3) Environmental Education curriculum calls for first hand study of the environments in the community.
4) Environmental Education should encourage investigations of local environment with the help of qualitative and quantitative techniques.
5) Environmental Education should be conducted in a scientific way with an open mind without being clouded by social and religious beliefs.



### 5:05:2 Infusing Environmental Education in the Existing Subjects

In this approach, the concepts of Environmental Education are infused in the existing subject curriculum, by relating them with the concepts of physics, chemistry, biology etc. As such, no new subject is added to the existing heavy curriculum but the concepts of Environmental Education being integrated with other subjects. The drawback with this approach is that the concepts of Environmental Education may be neglected with emphasis being laid on the concepts of the conventional and established subjects. As Environmental Education in this approach tends to become interdisciplinary, some of the teachers who are more familiar only with their subjects may find it difficult to give the appropriate treatment needed for Environmental

Education concepts. The teacher, in this approach, while discussing the concept of minerals will not expect the children to learn the names of minerals, but to explore the objects available or to make a list of the uses of minerals in everyday life. According to **Saxena** et al (1983), the following components in the curriculum have been infused according to the stage of mental development of the child:

1) Developing positive attitude towards environment
2) Relating the concepts with the local environment and resources
3) Attempting to develop emphathetic relationship with different members of the community and understanding their role and importance, and
4) Making use of easily available materials from the local environment for conducting experiments, demonstrations, discussions etc.

### 5:05:3 Occasional Programmes

In this approach, neither an extra subject is added to the curriculum, nor are the concepts of Environmental Education infused into the existing subjects. On the other hand some curricular or extracurricular activities are organized to take care



of the Environmental Education components. These activities or programmes, arranged on sequential basis or ad hoc, depending upon the requirements of the environment and the availability of time, money, and other resources, can be in the form of projects, excursions, trails, games, camps etc., and use of some educational material. These occasional programmes may last from a few weeks to a few months. **Bennet** (1982) provided 62 learning activities packaged into 6 units. Out of these, 24 selected activities make up a 9-week course. The learning activities are designed with the following features.

1) While incorporating behavioural objectives, the danger of relying solely on the theme to the exclusion of overlooking unanticipated outcomes, must be recognized.
2) A rich learning environment characterized by a competent teacher, efficient humanistic classroom organization and readily available resources must be encouraged.
3) A teacher-guided format of presentation, which is attractive, clear and motivating to the teacher, must be included.

4) Individual, small group and class instruction must be provided.

5) Teacher-directed, teacher-student directed, and student-directed learnings must be provided.

6) The use of primary and secondary sources of information must be incorporated.

7) Activities should be freely selected and sequenced into units, which meet local schedules, time constraints and special curriculum needs.

8) Provision must be made for the use of locally available instructional resources.

9) A variety of teaching styles should be reflected.

The activities were based on the following themes:

- Environment
- Social Life
- Economy
- Government

Emphasis was placed on the development of skills in inquiry, problem-solving, critical thinking, valuing, and psychomotor process. At the end of the course, the outcomes were measured in terms of achievement of social studies understanding and



skills. The advantage of the occasional Environmental Education programme is that the efficiency of the programme can easily be evaluated and established on the basis of pre and post-design. This will help the future users of the courses to choose a particular programme and identify the effect of the programme in a more clear-cut manner.

### 5:05:4 Environmental Education as Core Curriculum

In the normal practice of curriculum planning, each subject is independent of the other subjects, with each subject having its own textbook and curriculum. This approach may be regarded as 'atomistic' as disciplines form discrete and separate parts. In this atomistic approach, however, some basic problems arise when Environmental Education is incorporated in the curriculum. The main problem is that introduction of Environment Education results in less time for other subjects. This problem, of course, is there in other approaches too that are described above. Taking into consideration the problems of implementing Environmental Education in the school curriculum, **Fensham and May** (1979) suggested that Environmental Education should play the role of nucleus curriculum, as is being performed by science education at present. To develop the core curriculum, one may start with the characteristics of Environmental Education that form the basis of this core. Taking guidelines from the Tbillisi Conference, the following characteristics emerge:

- Environmental Education is oriented towards a problem.
- Environmental Education is concerned with realistic situations.
- Environmental Education aims at elaborating the alternatives that exist for situations and skills of choosing between them.
- Environmental Education includes action as an integral component.
- Environmental Education uses real environment of the school and its surroundings as a context.
- Environmental Education involves the clarification of values.
- Environmental Education aims at increasing the ability that students have to contribute to improving their own environmental situations.

With the help of these characteristics we can



identify how science and other subjects are contributing to Environmental Education.

The Approaches of Teaching Environmental Education may also be classified in terms of the varied emphasis in the curriculum, as follows:

- Emphasis on Skills
- Emphasis on Themes
- Emphasis on Environment as Medium of Learning
- Emphasis on Problem-Solving

## 5:06 Guidelines to Include Environmental Education at School Level

The following guidelines should be considered to incorporate Environmental Education at school level.

1. Environmental education at school level should aim at creating environmental awareness.
2. Environment concepts should be integrated in the existing courses in physical, natural and social sciences.
3. Students should be exposed to the concept of, nature education and the experience of participating in nature conservation programmes.

4. Various educational bodies should reorient early education curricula to include environmental issues.
5. Extra Curricular environmental programmes for school children are a powerful tool for imparting environmental education to children and should be increasingly encouraged.
6. Several techniques for providing extra curricular environmental education to school children are available. These include sending out by mail or distributing in the classroom, entertaining children's magazines on environment, organization of nature and science clubs, visits to exhibitions and museums, organizing painting and essay competitions and exhibitions, participation in community environmental action programmes, and using songs, plays, folktales, puppet shows and environmental games and puzzles.
7. At the higher secondary school level, it would be more purposeful to involve youth in seeking and finding solutions to environmental problems and thus inculcate a positive environmental attitude.
8. Distance education programme in environmental education for elementary



and secondary school teachers should be initiated.

9. Periodical workshops and seminars on environmental topics should be arranged for teachers.
10. Besides text books, other kinds of teaching aids for environmental education such as guides, handbooks, slide shows, films, models etc. should be developed.
11. Parent - Teacher associations should discuss environmental education strategies critically and evolve appropriate school-home integration for infusing the right attitude to environment in the children.
12. Programmes suited for non-school going and drop-out children should also be developed.

## 5:07 Current Status of Environmental Education in School Curriculum

Environmental education has been introduced in school curriculum at pre-school level, elementary level and higher secondary level both in State board (Tamil Nadu) schools and in Central (CBSE) schools. Both the State board (Tamil Nadu) syllabus and the CBSE syllabus are prepared in accordance with the National Policy on Education (1986).

### 5:07:1 Pre-school Level

At pre-school level, a general awareness about the personal hygiene and environmental cleanliness are introduced in a simplified version through a number of colorful diagrammatic illustrations. Rhymes related to nature are also introduced. The Government of Tamil Nadu has introduced **Scientific Tamil (Ariveyal Tamil)** at various levels in school curriculum, Environmental Education is one of the components included in the text book of Scientific Tamil. This book is published by the Tamil Nadu Text Book Society, Chennai - 6.

### 5:07:2 Elementary Level

At elementary level, the focus is stressed towards the environmental cleanliness. The concept of 'environment' is introduced. The relationship between the child and environment is emphasized. The child understands that he or she is surrounded by land, water, air, plants and animals. The role of environment is brought out through story telling and singing songs. But less emphasis is given to various practical activities in connection with environmental education.



### 5:07:3 Secondary Level

At secondary level, both in CBSE pattern and in State board School (Tamil Nadu) pattern the course materials are built on fundamental understanding of ecological and biogeographical principles. The important objectives are, to understand the ecological principles and issues, and to know abut the conservation. At the VI standard level, the environmental hygiene is explained in Scientific Tamil text book. The following lessons on environmental education are introduced in the Scientific Tamil syllabus at sixth standard level :

i) Primary energy

ii) Water harvest (through song)

iii) Environmental hygiene

iv) Medicinal plants

v) Life without diseases

Similarly in the VII standard syllabus of Scientific Tamil the following concepts related to environmental education are introduced:

i) Protection of soil, tree (through song)

ii) Electricity from clouds

iii) A search for planets

iv) Noise pollution

v) Chemicals in our daily life

vi) Food as medicine

But in the text book of Scientific Tamil these problems are not dealt in detail.

In science subjects some of the physical and biological components of environment are included. In history and civics the cultural, social and political components of the environment are added.

The National Council of Educational Research and Training (NCERT), New Delhi plays a pioneering role in the country in disseminating ideas about environment, environmental crisis and problems of population growth in a number of ways. Curricula for schools designed by the NCERT have centered on the local environment for lower classes while for the advanced grades, interrelated principles and global environmental issues have been brought to the fore. These have had a remarkable influence on the curricula developed by the different states of India. The textbooks prepared by the Department of Education in Social Sciences, Science and Mathematics reflect a broad spectrum of environmental issues that is intended to be a part of



general education in India. The textbooks for the social sciences, in Geography for instance, discuss at length the renewable and non-renewable resources of the country and conservation and afforestation as policy and practice. In addition, efforts have been made to make pupils aware of the factors related to ecosystem, wildlife, misuse of resources and need for their recycling.

Emphasis on evolving concepts of environmental education has been made in various generations of science textbooks developed by the NCERT. The scope and sequence of issues related to environment have been highlighted in three successive generations of instructional materials developed by the NCERT for classes VI to X and XI and XII.

The text books in Biology stressed population explosion, food problem and conservation on the one hand and on the other food chain and food webs and the biosphere as a whole were discussed in detail.

For lower classes, the material is developed in a manner that helps learning through environment. The idea behind this approach is that since the environment and the experience of children outside the school vary from place to place, the activities prescribed in the formal curriculum should be so designed that the experience are drawn from the environment of the child. The teachers' guide for classes I and II include the general philosophy and approach of environmental studies. The curriculum of environmental studies for classes III and IV has been designed with the objective of making the learner familiar with social and biophysical environments.

For the middle, secondary and higher secondary levels, human dependence on nature, adaptability of animals and man to their environments, population, ecological succession and eco-crisis, population and conservation, including community health, have featured in a well thought-out and coordinated way.

### 5:07:4 Higher Secondary Level

In general, at higher secondary level, only those students selecting science subjects like mathematics, physics, chemistry, and biology are exposed to environmental education. They are all more concerned with pollution ecology, population ecology and the role of science and technology in eliminating



or minimizing the various environmental problems. The arts students are inadequately exposed to environmental education.

## 5:08 Problems Faced in Imparting Environmental Education

Some of the major problems facing Environmental Education in the country can be described as follows :

***(i) Resource Constraints :*** Lack of resources is one of the major problems that is being encountered in the promotion of Environmental Education in the country. This is true of other developing countries too. It is a fact that any Environmental Education programme requires adequate resources, both in terms of money and personnel, if it is to be implemented successfully.

***(ii) High Dropout Rate :*** Because of the high dropout rate in our education system, teachers are left with no option but to begin the Environmental Education component from the primary classes to ensure that the students, even if they drop out later, are sensitized to the environmental problems.

***(iii) Social Constraints :*** Sometimes conclusions drawn from the study of Environmental

Education may clash with the prevailing social, religious and political thinking. This clash in thinking may lead to undesirable confrontation too. Environmental Education has no meaning and purpose if it is not accompanied by action. Ensuring this action is not easy.

***(iv) Difficulty in Assessment :*** Assessment of the work done as part of Environmental Education is difficult as many a time it is difficult to think of a common yardstick to evaluate the work done under different projects.

In effect, formal environmental education faces the following difficulties:

- A shortage of trained education officers in environmental education in the government to plan, organize, implement and monitor environmental education programmes in schools.
- A shortage of supporting staff and office equipment at the National Environmental Education Centre.
- Shortage of environmental education materials for teachers and students.



- Weak monitoring system
- Lack of good communication networks, cooperation and coordination within and among the various governmental and non-governmental organizations.
- Lack of government priorities or political will for environmental policy.
- Lack of recognition of communication as a valuable tool for policy, in part because of a misconception about communication as pertaining only to media, rather than as a relation with different parts of the policy cycle, and also because of competing demands for financial resources.
- Lack of appropriate legislative framework, enforcement and institutional support.
- Limited resources like funds and suitable materials to address the environment and environmental education.
- Government control of mass media.
- Lack of commitment and involvement on the part of the people concerned.
- High turnover of staff in schools and government.

- Lack of focus on schools and emphasis on formal education.
- Excessive curriculum and lack of teacher training and resources.

## 5:09 Important Methods of Instruction in Environmental Education

### 5:09:1 Discussion Method

A discussion is an open forum in which learners can express their opinions as well as review factual materials. In addition, discussion is a natural opportunity for students to exercise their command of the processes of communication, inference and conclusion (Wonfinger, 1984). In the discussion method the teacher gives a brief introduction of the topic for discussion. This is followed by supervised study by the students in groups or individually for an hour or so. The references are given by the teacher. The students may study in the class or may be allowed to go to the library for reference. The students can get their difficulties removed from the teacher. After the scheduled time the teacher initiates discussion by posting some questions or problems. By putting some key questions or problems in logical sequence the topic is covered through discussion.



Whenever, the discussion method is followed the teacher should keep in view the following points which are highlighted by Sharma (1996):

i) The topic for discussion should be chosen with due care and thought. It should commonly be of general nature – neither very simple nor very technical but which involves some thinking and interpretation on the part of the pupils.

ii) It should be made very clear to the students that they have not only to study the whole topic but have to read more, even from other sources about the part they have been assigned. To motivate them more the teacher may even say that he will see which group shows its best during the discussion.

iii) In order to involve maximum participation the teacher should see the discussion is not dominated by one or two students. He can even make a condition that every student will, at least, have to put one question and answer one question. In other words, no student should be allowed to put more than one question or answer one question at a time.

iv) Teacher has to be very careful that discussion is not stretched away from the topic but is relevant. Time should not be wasted for irrelevant discussion.

v) Healthy traditions of group discussion should be formed. More than one student should not be allowed to speak at a time. The arguments or view points of all the students are given due hearing and no one is ridiculed.

vi) Class discipline should not be disturbed.

vii) Any controversial point should be settled by the teacher at the proper time.

viii) The points left out by a particular group should be supplemented by the teacher.

## 5:09:1:01 Objectives of Discussion Method

The objectives of discussion method are listed hereunder:

i) To share information

ii) To clarify ideas

iii) To inspire interest

iv) To promote co-operative learning.

v) To identify the different views on a problem.

vi) To get conceptual clarity

vii) To develop the skill of expression



viii) To evaluate progress

ix) To locate and define the problem

x) To allocate responsibilities to find ways of solving the problem.

## 5:09:1:02 Types of Discussion

There are two types of discussion which are as follows:

i) Open Discussion

ii) Planned Discussion

### 5:09:1:02:1 Open Discussion

Open discussion is one in which the learner determines the topic and the role of the teacher is to ask questions that will lead the learner to consider various ideas. There can be no planning because the open discussion is spontaneous by definition. It can be extremely effective in getting learner to make inferences and draw conclusions. It is always on a topic of interest to the learners, because they initiate the discussion.

### 5:09:1:02:2 Planned Discussion

In a planned discussion, the teacher determines the content of the discussion, plans the questions, and guides the learners toward some predetermined goal. It is a way of introducing and teaching content to the learners in a way that will involve them cognitively.

Hence discussion is a natural opportunity for students to share their ideas freely with others in order to get conceptual clarity.

### 5:09:2 Seminar

The term 'seminar' is generally used to refer to a structured group discussion, that may proceed or follow a formal lecture, often in the form of extempore speeches or paper presentations. Individual students also prepare papers or report and present before a group of peers, as in the case of seminar paper presentation. The audience critically evaluates the paper and discusses the findings of the paper.

#### 5:09:2:01 Preparation for Seminar

The teacher should take the initiative in acquainting the students with the objectives and purpose of the seminar. Seminar requires much planning in terms of referring to literature on related aspects of the seminar topic, organizing the collected data in a sequential manner and presenting the paper through effective reporting. Duration of the



presentation of papers at seminars varies from topic to topic and discipline-to discipline. Generally 30-45 minutes are permitted for presentation followed by discussion for 10-15 minutes. Adequate time should be given to students or other participants to clear their doubts and probe the major aspects of the topic.

Observations by individual students appointed by the teacher, along with the teacher can be carried out in order to give a feed back to the presenter on his presentation. An observation schedule can be prepared in advance and training given to the selected observers to assess the presentation objectively and increase inter-observer reliability.

#### 5:09:2:02 Advantages of Seminar

i) The major advantage of the seminar is its stimulation and testing of students' power of comprehension and evaluation.

ii) The ability to detect and derive the principle from the context is developed.

iii) Understanding power and questioning ability in a relevant situation are strengthened.

iv) Self-reliance, self-confidence, sense of cooperation and responsibility are developed.

### 5:09:3 Workshop

Workshop is a get-together for some creative educational activity. While discussion demands much talk, workshop is a 'shop for work'. It is an activity-oriented technique. The group consisting of teachers, students, administrators, may initiate the workshop in a general session and frame guidelines for the conduct of the workshop. It involves directly the skills of both cognitive and psycho-motor domains. Preparing reports, syllabi, manuals and critical reviews, visiting places, making teaching-learning aids, and planning instructional designs, instructional materials and modules are examples of activities of a workshop session. Recording and reporting the work produced, at the concluding session is an important aspect of the workshop.

#### 5:09:3:01 Advantages of Workshop

The following are the advantages of workshop:

i) The workshop is based on the principles of learning by doing.

ii) It is an activity – oriented technique

iii) It is co-operative work which promotes the 'work culture'

iv) Teachers, students and administrators work together which ensure the sharing of ideas and views



v) It involves the skill of cognitive, affective and psychomotor domains

vi) A complicated problem can be easily solved in the workshop.

vii) The workshop focuses on the individual talents as well as the efforts of the group.

### 5:09:4 Field Trips

Field trips are very much educative and they create great curiosity in students and also bring out their creativity, **Keown** (1984) states that (1) those concepts that are integral part of the students' environment are best learned in the outdoor environment (ii) the concepts have a better chance of being understood and retained, if parts of the concept can be related to students' environment: (iii) critical thinking is enhanced in the outdoor environment: and (iv) investigations in the outdoor environment increase students' desire for that environment. The selected places for the field trips should be related to environmental problem. Before the execution of a field trip, the teacher has to plan the trip, including the teaching learning strategies to be followed. It is better if the teacher visits the place earlier and plans the field trip according to the resources available at the place. After the successful completion of the trip, based on the experience and observations, the students may submit their work in the form of reports, drawings, stories, poems etc. More field trips a teacher arranges, more awareness comes in the students towards environmental problems. **Lukeo** et al. (1982) concluded, based on three studies, that field trips is an important and effective method for clear cut cognitive and affective gain in the eyes of teachers and educators. The choice of environment and its setting play an important role in the formation of right attitudes and behaviours.

### 5:09:5 Field Surveys

Field survey is an analysis of the present status about a particular area. Through the field survey, one can get a correct picture of the status of a particular event. It is a natural observation. Rousseau emphasized that natural observation is the best method of teaching as compared with classroom teaching-learning and studying books. Through the field survey in environmental education, the learner gets first hand information about the environmental problems.



Students may study different aspects and objects of environment. They can study a village, a river, a lake, some specific problems of an area which include pollution aspects, inter relationship of living being etc., The outdoor studies also require a detailed planning like field trips, but these studies may be limited to the local environments. The study team has to plan the objectives of the study plan, execute and submit a report to take necessary action and for the benefit of other students.

#### 5:09:5:01 Types of Field Survey

Based on the objectives, the field survey can be classified into many types. Some of these are listed below:

i) Local field survey

ii) Regional field survey

iii) Geographical field survey

iv) Historical field survey

v) Industrial field survey

vi) Field survey of natural vegetation

vii) Field survey of animals

viii) Survey of soil and fauna

ix) Field survey of community

### 5:09:5:02 Objectives of Field Survey in Environmental Education

The following are the objectives of field survey:

i) To promote awareness about learner's environment

ii) To develop the tendency to survey it and utilise it for understanding.

iii) To develop the ability for interpreting the learners' own experiences and observations.

iv) To promote the skill of observation, and interpretation

v) To develop the ability of co-operation and group-work.

vi) To make familiar about the different tools and techniques that are used in the field survey.

vii) To develop the feeling of appreciation about the nature.

viii) To identify the qualities of the environment.

### 5:09:5:03 Steps Involved in Field Survey

The environment field survey has the following steps:

Step (i) Writing objectives

Step (ii) Planning



Step (iii) Identification of tools and techniques

Step (iv) Execution (Collecting information or data)

Step (v) Evaluation

Step (vi) Follow-up activities

Thus, field survey provides a vivid picture about a particular area. It is a scientific way of investigating the status of environment.

### 5:09:5:04 Enhancing the Effectiveness of Field Studies

**Backman and Crompton** (1984-85) feel that the effectiveness may be greatest if the outdoor experience is proceeded by an indoor experience which provides a cognitive framework into which pieces of information likely to be encountered in outdoor can be fitted. **Kirk** (1980-81) felt that (i) the activities which focus the attention on the use of nature study and field activities help the students to learn about the countryside and conservation; (ii) the activities which develop skills and interest in rigorous physical activities help in outdoor education (iii) the activities related to the study of man-made environment and social environment help students to learn urban traditions; and (iv) the activities related to the study of rural environment focus the attention of the students to learn about agriculture, horticulture, forestry and other forms of land management.

### 5:9:06 Projects

A Project is a problematic act carried to completion in its natural setting. The project method consists of building a comprehensive unit around an activity which may be carried on in the school or outside. ' Learning by doing', and ' Learning by living' are the two cardinal principles of this method. This method is psychologically sound. The teacher acts as a guide and helps the students to find the facts and principles themselves. The project may be a constructional type such as building up a school museum, running a school garden, collecting specimens for science exhibition or it may be the type meant for investigation such as growing seeds in a pot, the study of plant life or animal life.

The role of teacher is not of a dictator but a friend, guide and a working partner. The teacher should investigate along with the learner and should not claim to everything. Providing democratic atmosphere in the classroom is an important duty of the teacher. The teacher should be active all the time to see that the project is running on right lines.



In the project method, students have to take up certain projects. The total class is either divided into small groups or work at a total unit depending on the nature of the project to be worked out. Sometimes the project is also divided in to small units. Each student or small group of students is given a work for which the students take responsibility of completing it successfully. At the end of the project, the activities of all the students or groups are presented to the total class in the form of reports and displays. They explain their experiences, data, conclusions and recommendations. If we take the problem of mosquitoes as a project, a small group will identify the causes of mosquitoes, another will map the ditches, another will look into the eradication techniques, and another group will apply the selected techniques to eradicate the mosquitoes.

#### 5:09:6:01 Steps in Project Method

Steps involved in project method are as follows:

Step (i) Sensing a problem

Step (ii) Defining the problem.

Step (iii) Selection of appropriate methodology

Step (iv) Data Collection

Step (v) Process of analysis

Step (vi) Drawing Conclusion

Step (vii) Evaluation

Step (viii) Recording

## 5:09:6:02 Characteristics of a Good Project

The following are the characteristics of a good project:

i) Project should allow the active participation of both learners and teacher

ii) It should be useful and purposeful

iii) It should have definite educational value

iv) It should be practicable

v) It should provide maximum number of activities

vi) It should not be expensive

vii) It should promote the investigative ability among the learners.

## 5:09:6:03 Merits of Project Method

The following are the merits of project method:

1) This method is based upon the laws of learning i.e.
   a) Law of readiness
   b) Law of exercise
   c) Law of effect



2) It promotes co-operative activity and group-interaction

3) It is a democratic way of learning. The children choose, plan and execute the project themselves

4) It affords opportunity to develop keenness and accuracy of observation and to experience the joy of discovery.

5) It sets up a challenge to solve a problem and this stimulates constructive and creative thinking.

## 5:09:6:04 Demerits of Project Method

The following are the demerits of project method:

1) It absorbs a lot of time

2) It involves much more work on the part of the teacher.

3) The whole syllabus, especially for more advanced classes, cannot well be included in a collection of projects and it is difficult to finish the syllabus in the limited time.

4) Text books and materials written on these lines are not available.

5) It is expensive in the sense that a well-equipped library and a laboratory are required.

### 5:09:7 Exhibition

Exhibition in environmental education means presentation to view a display or showing off the materials relevant to environmental studies.

Exhibits or exhibitions can be arranged to show the project work of the students or to highlight the environmental problems in order to get suitable remedies. Different exhibits explaining various concepts of environmental education can be displayed in collaboration with various environmental organisations. Students will take part very actively in these exhibitions and show their abilities. They also explain the observers about the environmental problems and solutions.

The organisation of the exhibition should provide, for ample spheres for the development of skills of the members. The planning and the procedure should be detailed and should fulfill the ambitions of the members. As funds will be a constraint, adequate finance should be made available.

It is essential to form a committee to organize the exhibition. Work should be distributed among the various committees drawn for the purpose. The execution of the plans should be quick and



systematic. The teacher should guide the students in organizing the exhibition. Students should have conceptual clarity about the items that are to be displayed in the exhibition. There should be a spirit of healthy competition among the students to exhibit the materials. Moreover, teachers should be properly trained in organizing exhibitions. The leadership should gradually be passed from the teachers to students. The teacher should supervise the entire procedures. Evaluation of the items displayed in the exhibition should be done by experts in the respective fields.

#### 5:09:7:01 Advantages of Exhibition

i) Exhibition is based on the principle of learning by doing

ii) The learners can observe, analyse, criticize and apply the scientific laws.

iii) They get chances of picking up skills by means of participation in the exhibition

iv) They can learn how to co-operate and show leadership qualities while working in the exhibition.

v) Exhibition promotes scientific attitude among the learners.

vi) It helps the learners to use science in life situations.

vii) It promotes exploration and creative spirit among the learners.

viii) It propagates scientific information.

ix) It enhances the healthy spirit of competition among the participants.

x) The spirit of inquiry can be developed by organizing exhibitions.

### 5:09:8 Videos

Radio, tape recorder, television, Videos, video disc, computer, internet etc. are the non-print media that can be effectively utilized in the field of environmental education.

Videos form a valuable medium which helps learners to understand the abstract concepts. By means of video-show it is possible to bring the reality into the class room as it provides for visual stimulation through colour and motion. Moreover, it paves the way for individualized instruction in the class room. The learner can interact freely with the teacher when they are seeing a particular programme in a film or video-show. Specific concepts can be presented and discussed with a small group of students.



Sometimes it is very difficult to bring all the learners to the field in order to investigate a particular environmental problem. For example, to study the Narmada Controversy, it is very difficult to bring the learners to the particular site. Moreover it consumes a lot of time and it is not economical. Hence the teacher can arrange for a video-show about the Narmadha Controversy, by which the learners can easily understand the problem.

A documentary film can also explain ecological problems that happened in the past. For instance, in March 1973, in Japan it was brought into the notice that Chisso Corporation's aceto-aldehyde plant discharged aceto-aldehyde and mercury waste into the sea, where it got into the fish. These fishes were eaten by the local people resulting in loss of hearing, speech and sight and even death. This situation can be well illustrated with the help of the documentary film which is available in the market.

Many complex ideas can be explained by means of videos and films. It provides clarity and concreteness. Moreover it stimulates interest of the learners and retain their attention in learning.

### 5:09:9 Television

Television is a powerful medium which educate the learners and can develop a sense of participation. The T.V. has influence on the young generation and has access to all people. Technologically, as a combination of sound and pictures, it provides an effective communication to all. It is useful for mass education through satellite. The Educational Television (ETV) is a system that present learning content in various subject areas through programmes prepared by a central agency. Gyan Darshan, DD Kisan and Kalvi TV are telecasting the educational programmes.

#### 5:09:9:01 Advantages of Television Programmes

i) The Television Programme provides high quality of instruction.

ii) It promotes social equality for education.

iii) The learners learn from T.V. with their own efforts. Hence it reduces the dependency on teacher.

iv) The rapid changes in the curriculum and methods of instruction can easily be incorporated in educational T.V. Hence it ensures flexibility.

v) It provides education throughout the country at minimum cost and hence it is highly economical.



vi) T.V. has the advantage of the audio as well as the video aid.

#### 5:09:9:02 Limitations of Television Programmes

i) Teaching and learning through T.V. is an individualized method and thus co-operation, adjustment, cordial relationship etc. are not developed in viewers.

ii) The learners can only see the demonstration in the T.V. screen and thus they have no chance to handle it.

iii) The learners are the passive observers and are not active participants.

iv) The screen is small and the focused screens are not clear enough for large size class rooms.

v) The most important difficulty faced by institutions is to fit its academic schedules to the programme schedules of television stations.

### 5:09:10 Other Methods

**Simulation and Games**

Simulation and games can be used to focus attention on both attitudes and content. The advantage of games and simulations, according to Troost and Altman (1972) is that they have intrinsic potential for motivation.

### Debates

By arranging debates the teacher can make students aware of environment, its problems and the necessary feasible solutions.

### Readings

A teacher can ask the students to get further information through additional readings. This will help to grow individually.

### Inquiry

On finding a problem or else by a student of any occasion a student can take up an inquiry to probe into it. Or a teacher can also assign inquiries into various aspects of environmental education. The teacher should develop Inquiry Guides for the benefit of the students.

### Guest Lectures

Guest lectures given by eminent personalities will motivate the students in many ways and help the students to participate in environmental activities.

## 5:10 Environmental Education Programmes to be Undertaken in Educational Institutions

Mere knowledge about environmental problems does not develop the necessary behaviour in any individual to conserve his environment and its quality



(Lucas 1980). It is important to develop in pupils the requisite interest in solving the environmental problems. Following are some of the important ways and means suggested to develop the necessary values and feelings of concern for the environment,motivation for actively participating in environmental improvement and protection.

### 5:10:1 Arranging for Video-Clips

Video-clips on lusturous water falls, and mountains, dense forests and deep seas and majestic rivers of India, could be caught in digital cameras and the video-albums so prepared could be made available in the school 'Educational Technology Laboratory' for the students to enjoy and love their mother land and its vast natural resources.

Copies of photographs taken by students during their school excursion could also be compiled as an album and kept on view.

### 5:10:2 Establishing Environmental Club

As we have 'Science Club', 'Maths Club' etc. in schools, we could also form an 'Environmental Club' in which those interested in protection and conservation of environment are to be admitted as active members. Teachers, particularly science teachers should take the initiative in forming this club and charting out its functions.

### (i) Taking of Oath

Every member of environmental club should be urged to take an oath along with others in their class like "I, all through my life will not indulge in any activity that may affect and injure the environment".

### (ii) Field Trips

Teacher can take the students on a field trip to those places which have been subjected to severe environmental stress so as to gain first hand knowledge. [E.g. Industries which heavily discharge smoke and dust in the air, tanneries, shrimp firms, dyeing units, denuded hilly slopes, dry rivers, rivers that turned as sewage drains (such as the Coovam of Chennai, 'Odampokki' in Thiruvarur etc.) and paper industries].

### (iii) Celebrating the World Environmental Day

On behalf of the school environmental club, 5th June of every year could be observed as **"World Environmental Day"** by planting a tree-sappling in the school campus and distributing tree sapplings freely to the interested general public.

### (iv) Creating an Environmental Corner

At a vantage point in the school could be established an 'Environmental Corner' in which a display board, containing the latest information on



various environmental issues may be maintained. Teacher can encourage students to write interesting news on this display board as illustrated below:

★ In Chennai a 60 year old man has planted more than 1000 trees on the roadside and is watering them regularly.

★ In Erode District, a villager has removed single handedly the weeds spread on the river bed to a distance of 800 metre stretch and cleared the canal-path.

★ The Uttar Pradesh Government has created a 'Forest Grave' in which facilities are provided for people to plant tree-saplings in memory of their deceased kith and kin.

★ Getting permission from the Governor is made mandatory in Delhi, to fell trees

### (v) Tree Planting

Every one is to be encouraged to plant a sapling either in the school or at home on his/her birthday. Similarly pupils are to be urged to present a plant or tree-saple as birthday gift.

### (vi) Conducting Environmental Festival

Every year school children can conduct the environmental festival in a nearby village by arranging for cultural programmes, debates, dramas, and other art forms on themes related to environmental problems.

**(vii) Conducting Competitions in Environmental Awareness**

Annual competitions could be arranged for school children in environmental awareness. Students are to be encouraged to participate in competitions like 'poster making', 'drawing cartoons', 'essay writing', composing poems', 'elocution contest', 'group discussion' etc.

**(viii) Long Walk to Promote Environmental Awareness**

Students could be encouraged to undertake a 'Padayathra' (long walk), cycle rally or peace march to promote environmental awareness on the eve of 'World Earth day' and 'World Forest-Day'. While marching, students can carry playcards with catchy slogans like "Cigarette, the Slow Poison", "Each one, Plant one", "You smoke, I cough", "Let us Save Rain Water, the bliss of Heaven" etc.

## 5:10:3 Celebrating the World Environmental Day

5th June of every year should be celebrated in schools as the 'World Environmental Day'. Pupils could be encouraged to discuss on that day the environmental hazards that pose a grave danger to the whole of mankind.



That day could also be observed as **"Vanamahotsav"** day by planting tree saplings in and around the school. Competitions could be conducted in environmental awareness. Students are to be encouraged to participate in competitions like 'poster making', 'drawing cartoons', 'essay writing', 'composing poems', 'elocution contest', 'group discussion', etc. on environment related themes. Cultural programmes, debates, dramas and other art forms on themes related to environmental problems could be undertaken in the school with the involvement of the general public and parents. Each student can be made to take an oath on that day along with their friends, like "We all through our life will not indulge in any act that may affect and injure the environment".

## 5:10:4 Programmes on Hygiene and Sanitation

**i) Personal Hygiene**

Personal hygiene includes the following :

- Keeping the teeth, mouth, nails, limbs, ears, hair and the body clean and neat, by daily washing and brushing.
- keeping all the food stuffs and eatables well covered.
- not allowing water to stagnate near our house.

- always using protected water for drinking (well boiled and cooled water is preferrable)
- avoiding the habit of buying all sorts of edibles of our liking from the road side shops and eating at odd hours.
- washing hands everytime before and after eating.
- sleeping in well ventilated rooms.

**ii) Public Hygiene**

This may include the following :

- Solid wastes such as garbage, waste food, building scraps etc. should not be heaped on the roadsides; instead they are to be removed by using the garbage cleaning facilities provided by the civic authorities like the Municipality / City Corporation.
- refraining from urinating and defecating in public places.
- avoiding sneezing or spitting in public places.
- ensuring periodical vaccination of the pet animals brought up at home.
- keeping the home clean and attending now and then the sanitary needs of the toilets and bathrooms.

**iii) Sanitation**

This may include the following :



- not polluting the atmospheric air by burning solid wastes, like garbage, tyre etc. or in setting on fire-crackers on the roads / public places.
- not sounding the horns of vehicles loudly, particularly near hospitals and schools.
- avoiding loud speakers as far as possible, particularly in the night during festivals and celebrations.
- not throwing plastic wastes in the drains.
- not throwing litters on the roads.
- not letting out water from the sumps to flow on the road.
- avoid mosquitto menace by keeping closed the overhead water tanks and wells in our houses; not allowing sewage to stagnate in the drains.
- helping to maintain public toilets clean and hygienic.

Eduational institutions should periodically conduct programmes on hygiene and sanitation so as to develop proper awareness among the students, parents and the public and to follow hygienic practices and maintain public sanitation.

## 5:11 United Nations Environment Programme (UNEP)

United Nations Environment Programme (UNEP) is an international environmental agency operating across the world with the support of United Nations Organisation (UNO). It coordinates its activities, assisting developing countries in implementing environmentally sound policies and practices. It was founded by **Maurice Strong,** its first director, as a result of the United Nations Conference on the Human Environment in June 1972 and has its headquarters in **Giriri** neighbourhood of **Nairobi,** Kenya. UNEP has six regional offices and various country offices.

### 5:11:1 Functions of UNEP

Its activities cover a wide range of issues regarding the atmosphere, marine and terrestrial ecosystems, environmental governance and green economy. It has played a significant role in developing international environmental conventions, promoting environmental science and information and illustrating the way those can be implemented with the help of policies of governments and Non-governmental organisations. UNEP has also been active in funding and implementaing environment related development projects.

UNEP has aided in the formulation of guidelines and treaties on issues such as the international trade in potentially harmful chemicals, transboundary air pollution and contamination of international waterways.



UNEP, together with the World Meterological Organisation established the **Intergovernmental Panel on Climate Change** (IPCC) in 1988. UNEP is one of several **Implementing Agencies for the Global Environment Facility (GEF)** and the Multilateral Fund for the Implementation of Montreal Protocol and it is also a member of the United Nations Development Group. The International Cyanide Management Code, a programme of best practice for the chemical's use at gold mining operations, was developed with the help of UNEP.

## 5:12 Centre for Environmental Education (CEE)

The **Centre for Environmental Education (CEE)** was established in August 1984 with its headquarters in Ahemedabad as a **Centre of Excellence** supported by the Ministry of Environment and Forests. The organisation works towards developing programmes and materials to increase awareness about the environment and sustainable development. The Centre has 41 offices across India including regional cells in Bangalore (South), Guwahathi (North East), Lucknow (North), Ahemedabad (West), and Pune (Central); State Offices in Delhi, Hyderabad, Raipur, Goa and Coimbatore; and several field offices. It has international offices in Australia and Srilanka. **Dr. Ashok Khosla** is the present Chairman of CEE.

### 5:12:1 Functions of CEE

CEE has inherited rich multi-disciplinary resource base and varied experiences of **Nehru Foundation for Development,** its parent organisation, which has been promoting educational efforts since 1966 in the areas of science, nature study, health, development and environment.

At the time it began its activities, CEE was perhaps the only organisation actively engaged in environmental education in the country. It mainly aimed at creating environmental awareness in the communities, conducting widespread environmental education and training programmes through a very vast network. It has a vast range of Publications, books, posters, educational packages, bibliographies and directories. There is also a large computerised data base - the environmental education bank, which has a collection of more than 800 environment concepts, about 2500 environment related activities and hundreds of case studies.

In 1994, it was decided to move from environmental education to environmental action; As a result it began more pilot, field level and demonstration projects towards sustainable development which could be scaled up and replicated. Within ten years, these projects formed a major chunk of CEE's activities.



Today CEE is the nodal agency for the implimentation of UN Decade of Education for Sustainable Development (DESD) in India under the Ministry of Human Resource Development, Government of India. CEE's programmes in the decade will focus on 'Training and Capacity Building', 'Internships and Youth Programmes, Consultancy Services, Knowledge Centre for Education in Sustainable Development and **Journal on Education for Sustainable Development'.**

### 5:12:2 Thrust Areas of CEE's Work

1. Education for Children
2. Environmental Education at the Higher Education Level
3. Examination systems in Environmental Education.
4. Education for Youth.
5. Communicating environmental information through the media.
6. Experiencing Nature.
7. EE through interpretation
8. Knowledge Management for Sustainable Development
9. Industry initiatives
10. Sustainable Rural Development
11. Water and Sanitation

12. Sustainable Urban Development
13. Waste Management
14. EE for Fragile Areas
15. Biodiversity Conservation
16. Eco Tourism
17. Disaster Preparedness and Rehabilitation
18. Training, Networking, and Capacity Building
19. Facilitating NGOs and Community Initiatives
20. Initiatives for UN Decade of Education for Sustainable Development (DESD)

## 5:13 Role of N.C.E.R.T in Environmental Education

For the promotion of Environmental Education at school level NCERT undertakes the following activities.

1. Extension activities such as publication of quarterly journal 'School Science' and organizing a National Science Exhibition every year. The Themes of the exhibition are generally environment based.
2. Training Programme for teachers and teacher educators.
3. Development of films/ video programmes on Indian environment. There is a film library in NCERT that loans films to schools.
4. Publication of supplementary reading materials.



5. Collaboration with international agencies like UNESCO, UNICEF, UNEP etc.

## 5:14 Role of Teachers in Environmental Education

The objective of environmental education is to learn the skills of gathering information, developing desirable attitudes, values and habits. Hence the teacher has to act as a guide and stimulate the children to use their abilities to acquire information.

The teacher may structure the learning experiences and activities and prepare the points of enquiry. The students can be divided into small groups to conduct the investigations. The groups may exchange their experiences in the general class and enter into discussions. The functions of the teacher may be summarized as follows:

a) to arouse the children's interest in the environment and to raise challenging problems;
b) to discuss the approach to problems or topics;
c) to organize working groups and to provide with the help of work cards for the lines of enquiry;
d) to arrange visits or expeditions;
e) to provide reference materials for children's use;
f) to provide materials needed for practical work;
g) to arrange for visiting speakers;
h) to initiate and develop discussion and debate; and

i) to provide facilities for displays and exhibitions of the work carried out.

## Conclusion

In this unit, status of environmental education in school curriculum, incorporating environmental education in the school curriculum at different levels of formal education, innovative methods of teaching environmental education, problems faced in teaching environmental education, role of UNEP, CEE and NCERT in promoting environmental education, were discussed in detail.

## Review Questions

1. Explain the contributions of National Policy on Education (1986) (B) [Ans: 5:01]
2. What is the need for Environmental Education in Schools? (B) [Ans : 5:02]
3. Briefly explain the materials of environmental education for different stages of formal education. (B) [Ans: 5:03]
4. Discuss the various approaches followed in environmental education. (A) [Ans: 5:04 + 5:04:1 + 5:04:2]
5. Discuss the different ways of incorporating the content of environmental education in the school curriculum with the merits and limitations of each one. (A) [Ans. 5:04 + 5:04:1 + 5:04:2 + 5:05 + 5:05:1 to the end of 5:05:4]



6. Discuss the current status of Environmental Education in Indian school curriculum at different levels. (A) [Ans: 5:07 + 5:07:1 to the end of 5:07:4]
7. Briefly discuss the problems faced in imparting environmental education. (B) [Ans: 5:08]
8. Briefly explain the following:
   i) Seminar method (B) [Ans: 5:09:2 + 5:09:2:01 ]
   ii) Workshop method (B) [Ans: 5:09:3 + 5:09:3:01]
   iii) Field Survey (B) [Ans: 5:09:5 + 5:09:5:01 to the end of 5:09:5:03]
   iv) Discussion Method (B) [Ans. 5:09:1]
   v) Field Trips (C) [Ans. 5:09:4]
9. What are objectives of Discussion Method? (C) [Ans. 5:09:1:01]
10. Explain the two types of discussion. (C) [Ans. 5:09:1:02 and its 2 sub divisions]
11. What is a seminar? (C) [Ans: 5:09:2]
12. Mention the advantages of workshop (C) [Ans: 5:09:3:01]
13. What are the objectives of Field-Survey ? (C) [Ans: 5:09:5:02]
14. Mention the Steps involved in a field survey (C) [Ans: 5:09:5:03]
15. What are the characteristics of a good project? (C) [Ans. 5:09:6:02]
16. Mention the steps involved in a project method. (C) [Ans. 5:09:6:01]
17. State the merits and demerits of project method. (B) [Ans. 5:09:6:03 + 5:09:6:04]

18. Briefly explain the role of exhibition in the learning of 'Environmental Education' (B) [Ans: 5:09:7 + 5:09:7:01]

19. Briefly describe the programmes to be undertaken in educational institutions to develop environmental awareness. (A) [Ans: 5:10 + 5:10:1 to the end of 5:10:4]

20. Write a note on each of the following :

    i) Clean and green campus programme (B) [Ans. 5:10:4]

    ii) Environmental Club (B) [Ans. 5:10:2]

    iii) Video show on Environment (C) [Ans. 5:09:8]

    iv) Conducting Environmental Festival (C) [Ans. (vi) of 5:10:2]

    v) Celebrating the World Environment Day. (C) [Ans. 5:10:3]

    vi) Use of Television in Environmental Education. (C) [Ans. 5:09:9]

    vii) Presenting Video-Clips (C) [Ans. 5:10:1]

21. Briefly describe United Nations Environment Programme. (B) [Ans. 5:11 + 5:11:1]

22. Describe about Centre for Environment Education and its functions. (A) [Ans. 5:12 + 5:12:1 + 5:12:2]

23. Discuss the guidelines to be incorporated in environmental education at school level. (B) [Ans. 5:06]

24. Write a note on 'Role of NCERT' in Environmental Education. (C) [Ans. 5:13]

25. State the role of teacher in Environmental Education (B) [Ans: 5:14]

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# ENVIRONMENTAL EDUCATION

(Model Question Papers based on the revised syllabus of 2021 with answers cited in the book)

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